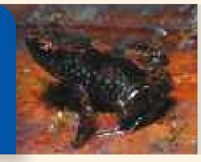
52 Chapter Review



Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT 52.1

Earth's climate varies by latitude and season and is changing rapidly *(pp. 1167–1170)*

- Global **climate** patterns are largely determined by the input of solar energy and Earth's revolution around the sun.
- The changing angle of the sun over the year, bodies of water, and mountains exert seasonal, regional, and local effects on climate.
- Climate affects where plants can live, but these effects run both ways: Vegetation can alter local and regional climate.
- Fine-scale differences in certain **abiotic** (nonliving) factors, such as sunlight and temperature, determine **microclimate**.
- Increasing greenhouse gas concentrations in the air are warming Earth and altering the distributions of many species. Some species will not be able to shift their ranges quickly enough to reach suitable habitat in the future.

? *Suppose global air circulation suddenly reversed, with moist air ascending at 30° north and south latitude and descending at the equator. At what latitude would you most likely find deserts in this scenario?*

CONCEPT 52.2

The distribution of terrestrial biomes is controlled by climate and disturbance *(pp. 1171–1176)*

- **Climographs** show that temperature and precipitation are correlated with **biomes**. Because other factors also play roles in biome location, biomes overlap.
- Terrestrial biomes are often named for major physical or climatic factors and for their predominant vegetation. Vertical layering is an important feature of terrestrial biomes.
- **Disturbance**, both natural and human-induced, influences the type of vegetation found in biomes. Humans have altered much of Earth's surface, replacing the natural terrestrial communities described and depicted in Figure 52.13 with urban and agricultural ones.
- The pattern of climatic variation is as important as the average climate in determining where biomes occur.

? *In what ways are disturbances important for savanna ecosystems and the plants in them?*

CONCEPT 52.3

Aquatic biomes are diverse and dynamic systems that cover most of Earth *(pp. 1177–1178)*

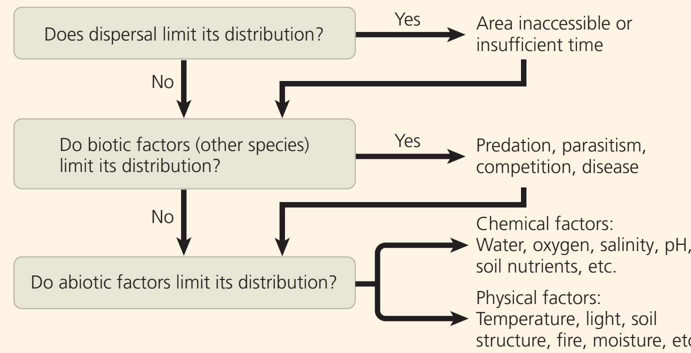
- Aquatic biomes are characterized primarily by their physical environment rather than by climate and are often layered with regard to light penetration, temperature, and community structure. Marine biomes have a higher salt concentration than freshwater biomes.
- In the ocean and in most lakes, an abrupt temperature change called a **thermocline** separates a more uniformly warm upper layer from more uniformly cold deeper waters.
- Many temperate lakes undergo a **turnover,** or mixing of water in spring and fall, that sends deep, nutrient-rich water to the surface and shallow, oxygen-rich water to deeper layers.

? *In which aquatic biomes might you find an aphotic zone?*

CONCEPT 52.4

Interactions between organisms and the environment limit the distribution of species *(pp. 1178–1187)*

Ecologists want to know not only *where* species occur but also *why* those species occur where they do.



VISUAL SKILLS *If you were an ecologist studying the chemical and physical limits to the distributions of species, how might you rearrange the flowchart preceding this question?*

CONCEPT 52.5

Ecological change and evolution affect one another over long and short periods of time *(p. 1187)*

- Ecological interactions can cause evolutionary change, as when predators cause natural selection in a prey population.
- Likewise, an evolutionary change, such as an increase in the frequency of a new defensive mechanism in a prey population, can alter the outcome of ecological interactions.

? *Suppose humans introduced a species to a new continent where it had few predators or parasites. How might this lead to eco-evolutionary feedback effects?*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Which of the following areas of study focuses on the exchange of energy, organisms, and materials between ecosystems?
(A) organismal ecology
(B) landscape ecology
(C) ecosystem ecology
(D) community ecology

2. Which lake zone would be absent in a very shallow lake?
(A) benthic zone
(B) aphotic zone
(C) pelagic zone
(D) littoral zone

Levels 3-4: Applying/Analyzing

3. Which of the following is characteristic of most terrestrial biomes?
(A) a distribution predicted almost entirely by rock and soil patterns
(B) clear boundaries between adjacent biomes
(C) vegetation demonstrating vertical layering
(D) cold winter months